CP Sheet 07

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We didn't have too much time this week and skipped some tasks which we considered less relevant – so you can be happy since you don't have to read too much this week...

Part b

After having implemented the bond interactions, we simulated for the given parameters and saved the trajectories into the corresponding files (see `main_simulationpy.sec` - or in `main_simulation.py` if Ilias finally showed mercy with us... ;)).
Hereby, we got the plot length probability distribution $P(b)$ as can be seen in fig. 1. The plots have been created in `main_exercises.py`.

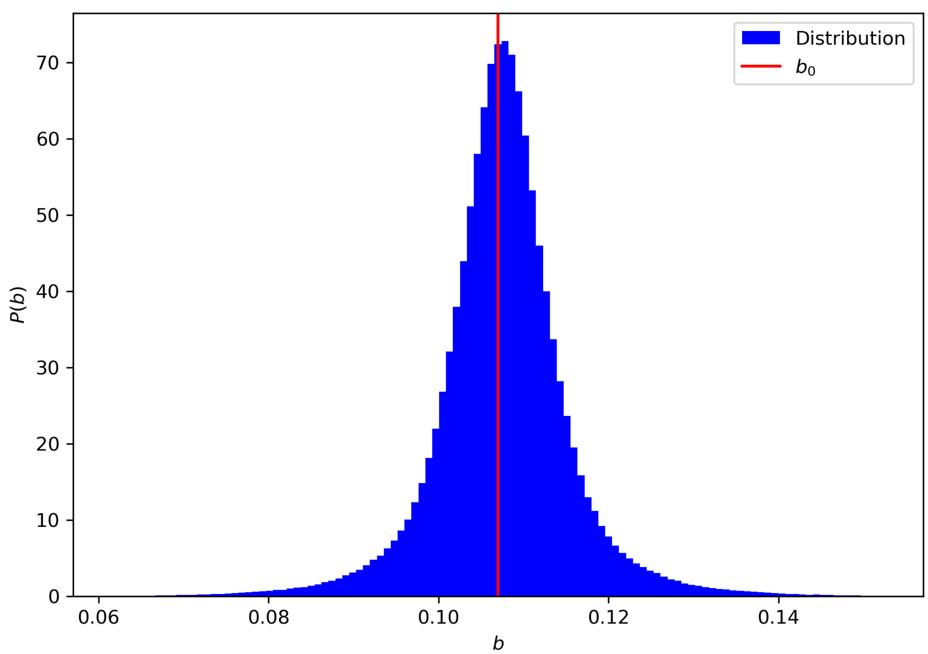

Figure 1: Probability distribution $P(b)$.

In fig. 1, a normal distribution of the bond lengths can be observed. The mean deviates slightly from the expected value b_0, which becomes even more obvious when looking at the time evolution of the mean bond length $\langle b(t) \rangle_N$ as shown in fig. 2. Here, one can also see that there is no drift but only fluctuations between a stable mean.
This deviation of the mean could be caused by the attraction of the other molecules (which dominates compared to the repulsion at the given density) which pulls the particles apart. This could be checked with more time by simulating other densities, similarly one might also vary the number of particles and thereby check for finite size effects.

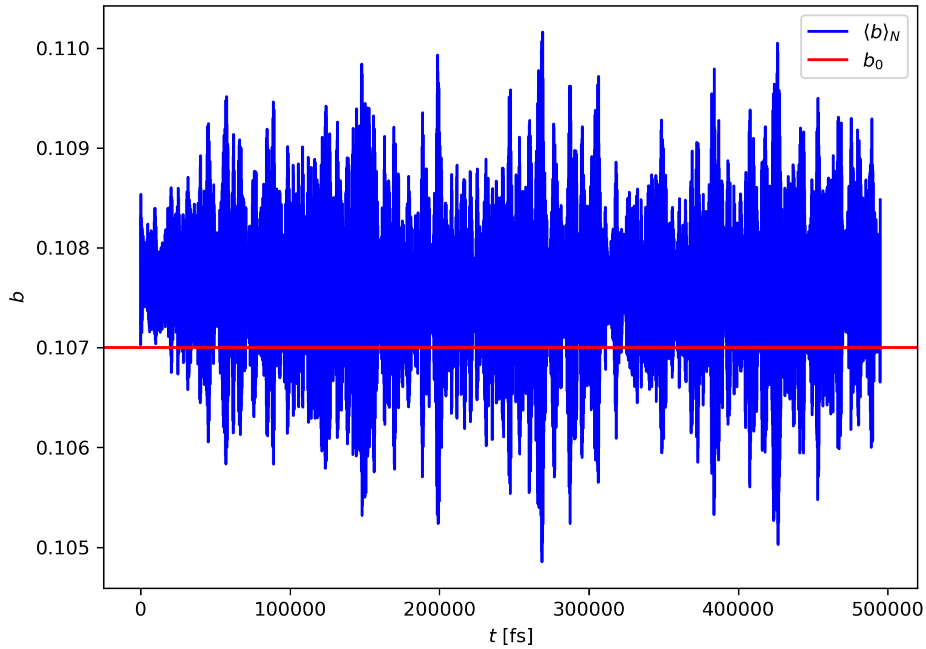

Figure 2: Time evolution of the mean bond length $\langle b(t) \rangle_N$.

We then calculated the time evolution of the average bond energy $\langle U_{\mathrm{bond}}(t) \rangle_N$ and compared to the expected value of $1/2\, k_B T = 0.298\,\mathrm{kcal/mole}$. This is displayed in fig. 3 where one can see that the average bond energy strongly fluctuates, but is in only slightly larger than the expected value which corresponds to the earlier observed larger mean bond length $\langle b(t) \rangle_N$.

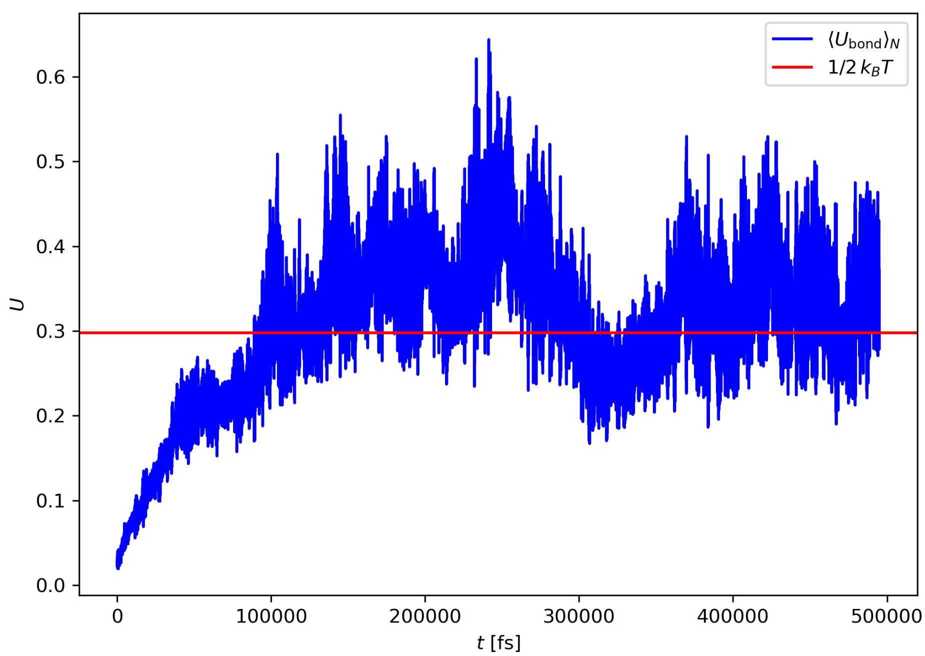

Figure 3: Time evolution of the average bond energy $\langle U_{\mathrm{bond}}(t) \rangle_N$.

Lastly, we computed the radial distribution function with
(i) r being the distance between the center-of-mass of distinct molecules and (ii) r being the distance between

individual atoms (as for a standard atomic RDF).
Both cases are displayed in fig. 4, where one can observe the behavior known from monoatomic simulations when calculating the $g(r)$ for the center-of-mass of distinct molecules. For the $g(r)$ measured for all atoms, one can additionally observe a high peak at a smaller distance then the first hydration shell, which corresponds to the intermolecular distance, i.e. the bond length. For large r, both converge to the same behavior.

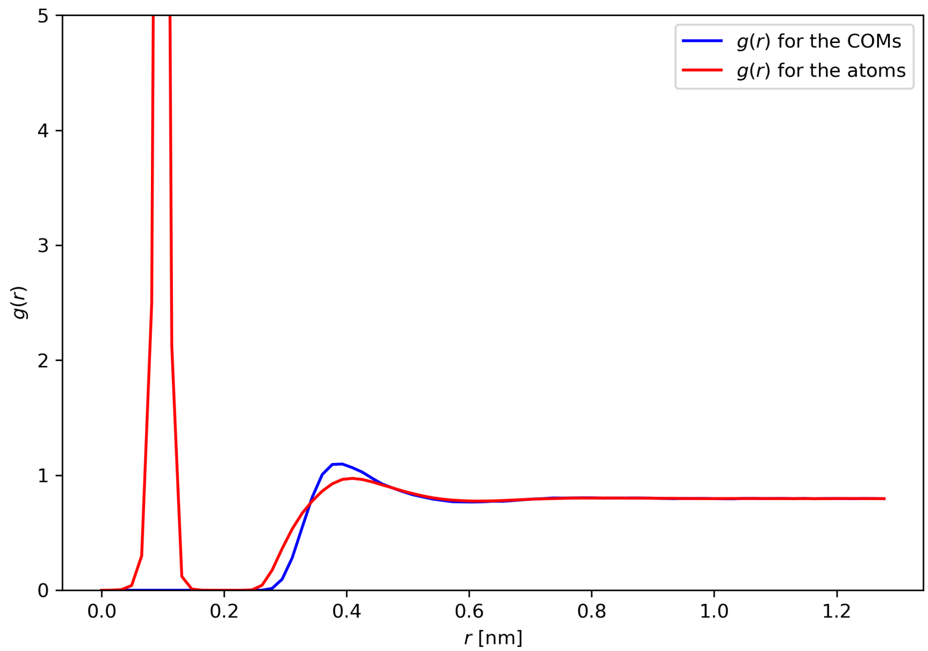

Figure 4: Radial distribution functions.

Additionally, we also provide a representative OVITO snapshot of the system at the end of the production run in fig. 5 where one can clearly see the individual molecules; some of them seem to be divided, but a closer look reveals that they are still connected through the periodic boundaries, as for each single particle, the corresponding partner can be spotted in the periodic neighborhood.

Part c

In part c and d, we analyzed the trajectories obtained in part b; in part c, we used the data generated for every time step since we are interested in the short time dynamics.

We first calculated the time–dependent autocorrelation function of molecular orientations $C(t)$, which is implemented in `main_exercises`. We then fitted $C(t)$ to a single-exponential decay model $C(t) \approx \exp(-t/\tau_R)$, which is visualized in fig. 6.
Here, the fit delivered the relaxation time $\tau_R \approx 268.258\,\mathrm{fs}$, which is much bigger than the characteristic vibrational timescale of a bond $\tau_{\text{bond}} = \omega^{-1} \approx 0.038\,\mathrm{fs} \ll \tau_R$.
This big difference can be explained by the fact that bond vibrations are fast, local oscillations caused by strong intramolecular forces, while orientational relaxation involves the much slower rotational diffusion of the entire molecule due to interactions with surrounding particles. Thus, the relaxation of molecular orientation happens on a much longer timescale than the internal bond vibrations.

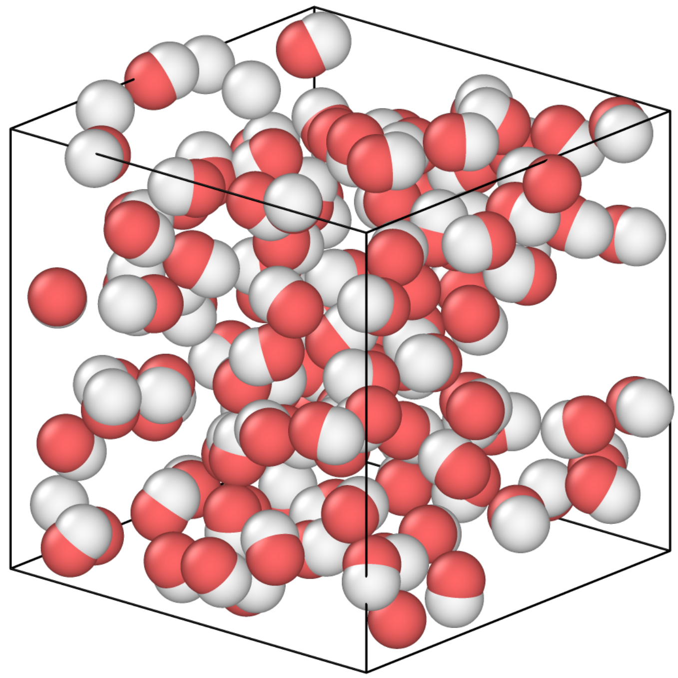

Figure 5: Representative OVITO snapshot.

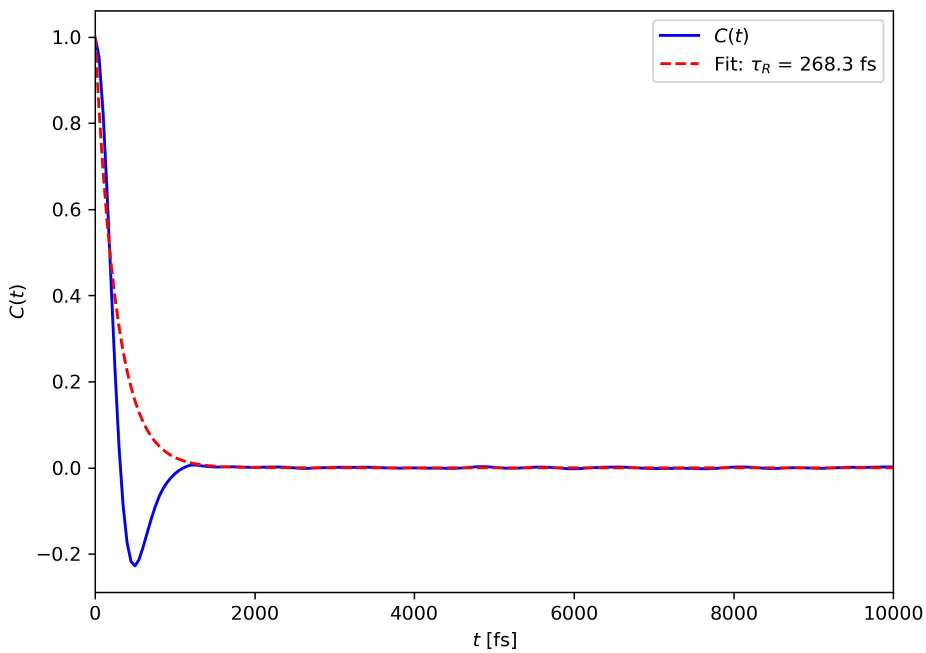

Figure 6: Time–dependent autocorrelation function of molecular orientations $C(t)$ fitted to a single-exponential decay.

Part d

Using the data for long time dynamic from part b, we finally calculated the mean–squared displacement (MSD) of the molecules' center–of–mass. Here, we first struggled with the fact that all particle positions from the trajectory file are reboxed, thus we first had to define a function to unwrap these coordinates. The implementation ads or subtracts the box length in the corresponding direction each time the particle crosses the periodic boundaries; it can be found in `main_exercises`, an exemplary application on the x–coordinate of one particle is shown in fig. 7.

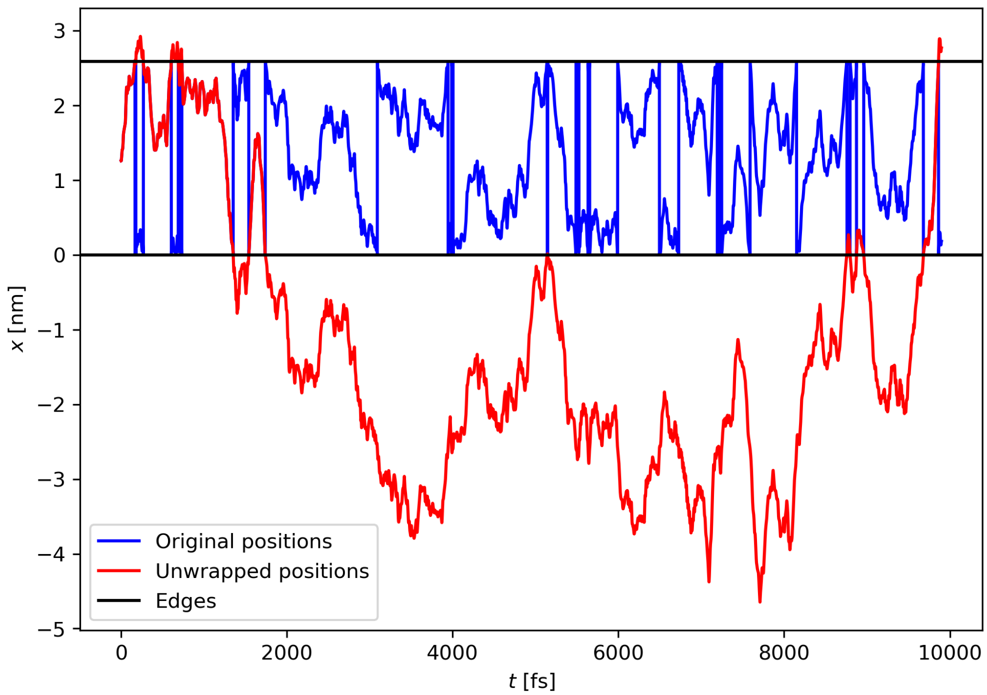

Figure 7: Unwrapping of particle positions.

Having obtained the unwrapped trajectories, we can now compute the MSD as displayed in fig. 8.
Here, one can observe a quadratic dependency (thus a slope of 2) in the double logarithmic plot for times smaller than $t \approx 10^3$ fs, which correspond to the ballistic regime. For larger t, the MSD converges more and more to a linear dependency on the time, shown by a slope of 1.

From the MSD, one can also compute the diffusion coefficient by applying the Einstein relation. Here again, we can observe the effects of the ballistic regime, since the "instantaneous diffusion constant" $D(t) = \mathrm{MSD}(t)/6t$ is instable in the beginning. The diffusion constant described in the Einstein relation is the limit of $t \to \infty$, but as seen in the plot for $D(t)$ in fig. 9, our simulated values for t are large enough for $D(t)$ to converge, so it is justified to interpret the last value $D = D(t = 0.495\,\mathrm{ns}) \approx 6.36 \cdot 10^{-5}\,\mathrm{nm}^2/\mathrm{fs}$ as the diffusion constant of the di–nitrogen molecules.

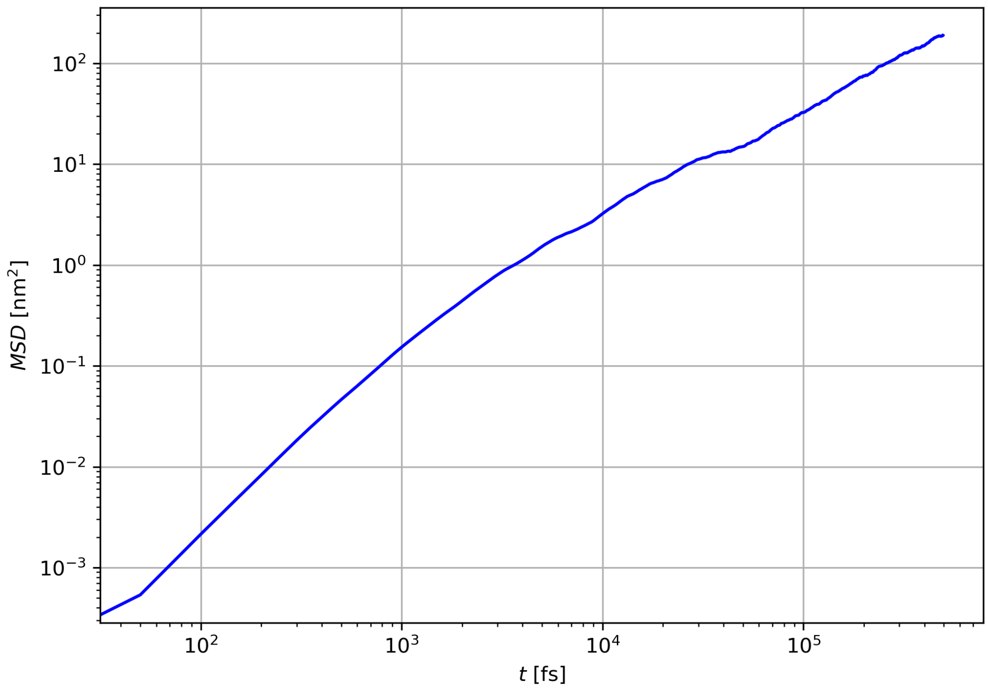

Figure 8: Mean–squared displacement of the molecules' center–of–mass in a double logarithmic plot.

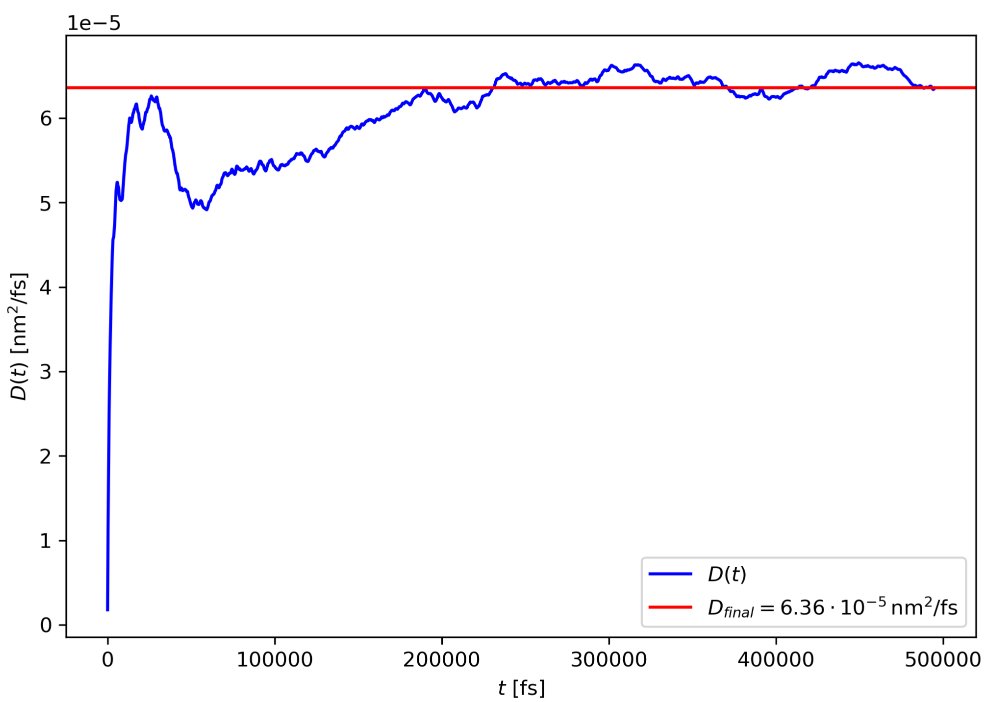

Figure 9: Time evolution of the instantaneous diffusion constant.